DO NOW — RECALL

Lesson 121 · Writing Algorithms with Validation

Name: _____ Class: _____ Date: _____

1. Why must a trace table record changes rather than only final output?

2. After pushing A then B onto a stack, which is popped first?

3. Why must a binary search operate on ordered data?

4. An OR gate has inputs 0 and 1. What is the output?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

It reveals execution path and when variable values change.

2. Model answer

B, because a stack is LIFO.

3. Model answer

It discards half the search space using an ordering comparison.

4. Model answer

1.

By the end, you can…

• State valid range, type and required behaviour before coding.

• Use sequence, selection and iteration with clear termination.

• Use normal, boundary, erroneous and extreme cases appropriately.

Vocabulary — know these terms

validation — Checking that input meets rules.

robustness — Ability to handle invalid/unexpected input safely.

iteration — Repetition based on loop condition.

boundary — Input at/near permitted limit.

NEW LEARNING

Read, annotate and explain

Define constraints

State valid range, type and required behaviour before coding.

Build control flow

Use sequence, selection and iteration with clear termination.

Prove with tests

Use normal, boundary, erroneous and extreme cases appropriately.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Write OCR-style pseudocode that repeatedly requests an age from 11 to 19 inclusive.

Model answer

```
age = int(input("Age"))

while age < 11 OR age > 19

age = int(input("Age"))

endwhile

print(age)
```

Note: this handles numeric range, not a non-numeric input error without extra handling.

Check: link each claim to the scenario and use precise OCR terminology.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Write an algorithm accepting a positive integer (at least 1) and outputting it. [6]

Question 2

Explain why 'WHILE age < 11 AND age > 19' fails as range validation. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Extend a validation algorithm to reject non-integer input, describe robust behaviour and supply three tests. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

When rejecting values outside a range, the two invalid inequalities should usually be joined with AND.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.